Data Clustering Using Decision Tree Algorithm for Business Intelligence

In [2]:
```
# Siddhi Pawar B-64
```

In [4]:
```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
```

In [6]:
```
# Load dataset
df = pd.read_csv('Decision_Tree_Dataset.csv')
```

In [8]:
```
df.head()
```

Out[8]:

	Initial_payment	Last_payment	Credit_score	House_number	Sum	Result
0	7528	7325	670	3253	12941	True
1	6183	2419	503	4395	20647	False
2	8859	2251	651	2419	15228	False
3	3406	5071	829	2225	15474	False
4	9793	1696	462	4824	11512	False

In [10]:
```
# Features and target
X = df.drop('Result', axis=1)
y = df['Result']
```

In [12]:
```
# Create model
model = DecisionTreeClassifier(
    criterion='entropy',     # or 'gini'
    max_depth=3,
    min_samples_split=2,
    random_state=42
)
```

In [14]:
```
# Train model
model.fit(X, y)
```

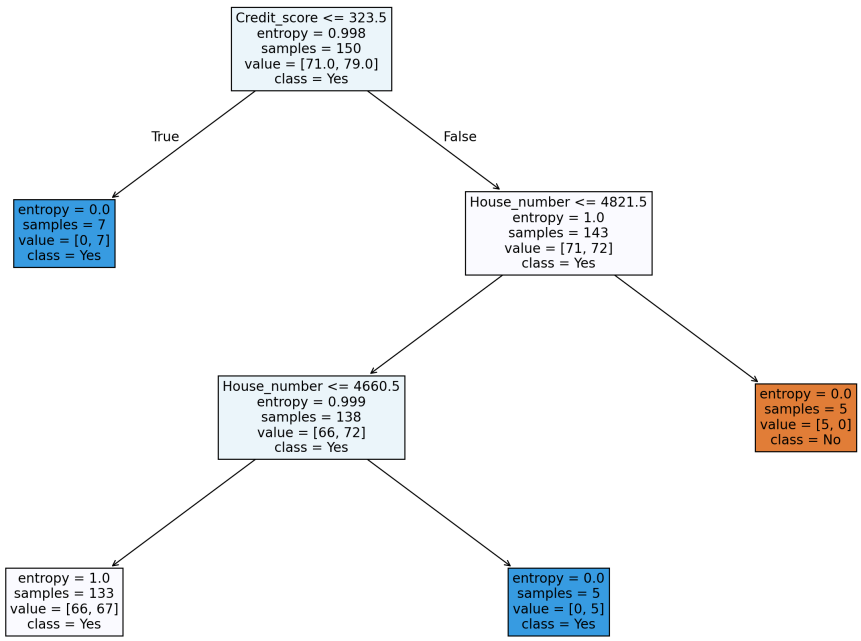
Out[14]:

```
DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', max_depth=3, random_stat
e=42)
```

In [30]:
```
# Plot tree
plt.figure(figsize=(20,12), dpi=200)
```

```
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['No', 'Yes'],
    filled=True,
    fontsize=12
)

plt.show()
```



In [ ]: